BENEFITS INTELLIGENCE AGENTS

Benefits Review Customer Feedback Implementation Report

Summary of the catalog, extraction, batching, model-configuration, validation, and alignment improvements implemented for the Singapore Benefits Review proof of concept.

Scope: Singapore Company and Market profiles
Catalog: GLB Benefit Datapoints Full V1 (Standard), 160 datapoints
Latest evaluated model: GPT-5.6 Luna, low reasoning
Prepared: 12 November 2026

1. Executive Summary

> The implementation now uses the authoritative GLB V1 catalog one-to-one, processes a maximum of 15 datapoints per model call, retains complete topic evidence in every batch, retries omitted keys, and reports model, parser, and aggregation losses separately. The latest Singapore Luna run returned all 160 requested keys with no missing placeholders and no parser losses.

160/160	0	15	1/1
Authoritative GLB V1 keys returned	Missing rows after processing	Bounded topic batches	Initially omitted key recovered by retry

The work materially improves catalog fidelity and structural completeness. Factual extraction validation remains the primary acceptance criterion. A known example in the latest run returned an explicit Short-Term Disability market statistic as `Not Provided`; this is being treated as an extraction-quality issue rather than a missing-key or parsing failure.

2. Complete Feedback-to-Implementation Traceability

The table below expands the consolidated themes into the individual feedback items raised across the catalog review and the subsequent three-part extraction, batching, and model review.

Customer feedback	Implementation completed	Status
The previous 160 output rows were not the same 160 datapoints as the original catalog; some appeared split, consolidated, or missing.	Converted `GLB_Benefit_Datapoints_Full_v1.csv` one-to-one into the runtime JSON catalog, preserving all 160 keys, prompts, and source order.	IMPLEMENTED
Test both Standard V1 and Simplified V2 with the same inputs and model before selecting a catalog.	Standard V1 is converted, deployed, and validated. The same converter/pipeline pattern is ready for V2 after the V1 factual baseline is established.	V2 PENDING
Prioritize Company Provision and Market Practice accuracy over alignment agreement.	Changed the system instructions so extraction accuracy and completeness explicitly take priority over alignment.	IMPLEMENTED
Use the golden dataset as a factual reference, not as an exact-wording or text-similarity target.	Defined factual evaluation around benefit characteristics, values, populations, denominators, and source support. Text similarity is not treated as the acceptance metric.	SCORING IN PROGRESS

Customer feedback	Implementation completed	Status
Prevent omissions of important benefit details.	Added a prompt-completeness check requiring every supported characteristic requested by the original GLB prompt, including formula, limit, eligibility, frequency, duration, conditions, and subgroup variations.	IMPLEMENTED
Prevent incorrect market statistics or denominator/population basis.	Added exact-market-basis rules that preserve each statistic with its option, population, and denominator and prohibit inferred, combined, rounded, reversed, or invented statistics.	IMPLEMENTED
Prevent evidence from related benefits being assigned to the wrong datapoint.	Added independent datapoint scope and nearby-evidence checks covering benefit name, characteristic, population, unit, and period before evidence can be assigned.	IMPLEMENTED
Make better use of the existing datapoint prompts.	Every original GLB prompt is retained exactly and sent as structured JSON, avoiding corruption from multiline text or pipe characters.	IMPLEMENTED
Limit each LLM call to a maximum of 15 datapoints, especially for dense Healthcare content.	Added a hard 15-datapoint batch limit. Healthcare's 47 datapoints are processed as `15 + 15 + 15 + 2`.	IMPLEMENTED
Confirm every Healthcare batch receives the complete and correct Company and Market sections.	Every internal batch reuses the complete relevant Company and Market topic slices; only the datapoint subset changes.	IMPLEMENTED
Retry model-omitted records before creating placeholders.	Each batch compares requested and returned keys, then retries only omitted datapoints once with the same complete topic evidence.	IMPLEMENTED
Determine whether records were omitted by the model or lost during response processing.	Added diagnostics for raw structured keys, model omissions, parser-dropped keys, retry recovery, and keys missing after aggregation but before placeholder fill.	IMPLEMENTED
GPT-5 reasoning models do not support temperature; omit it and configure reasoning effort instead.	GPT-5/Luna/Terra requests omit `temperature` and `top_p` and send reasoning effort. The latest Luna deployment used `low`.	IMPLEMENTED
Establish a GPT-4.1 baseline with smaller batches and test temperature 0 versus 0.2.	GPT-4.1 retains configurable sampling parameters and can run through the identical updated pipeline. Controlled baseline runs remain to be completed.	READY TO TEST

Customer feedback	Implementation completed	Status
Assess consistency when experimenting with a higher GPT-4.1 temperature.	The comparison plan includes repeated runs and separate measures for factual accuracy, consistency, omissions, denominator errors, and wrong-benefit attribution.	PLANNED
Run the same inputs, datapoints, and prompts through Luna and Terra at reasoning none/low.	Luna low has completed an end-to-end run. Terra was not present in the Foundry deployment inventory, and additional reasoning configurations remain pending.	PARTIALLY COMPLETE
Compare actual extraction quality, consistency, and performance to determine whether model choice is material.	The pipeline now captures complete result metadata and token usage, but a controlled model-only comparison against factual golden scoring is still required.	IN PROGRESS

3. Catalog Fidelity

Problem identified

The original runtime catalog also contained 160 rows, but those rows did not match the customer's authoritative 160 GLB keys. Equal row counts had obscured splitting, consolidation, and taxonomy differences.

Implementation

- The Standard V1 CSV is converted one-to-one; rows are not split or consolidated.
- `ComparisonKey`, prompt text, sort order, topic, benefit type, and source metadata are retained.
- The converter rejects blank or duplicate keys and enforces the expected count of 160.
- The deployed image identifies the catalog as `glb-v1-standard-2026-07-31`.
- The original GLB prompts are serialized as JSON, preventing multiline text and pipe characters from corrupting prompt boundaries.

> **Verified deployment signal:** the latest result begins with `risk_benefits_life_insurance_benefit_offered`, rather than an old custom key such as `ret_prev`.

4. Extraction Accuracy Improvements

The system prompt now makes extraction accuracy and completeness higher priority than alignment. It requires the model to:

- treat every comparison key as an independent extraction task;
- use only evidence applicable to the current benefit, population, option, unit, and period;
- include all requested, supported characteristics rather than stopping at the first matching fact;
- preserve market values together with their option, population, and denominator basis;
- avoid calculating, reversing, rounding, combining, or inventing statistics; and
- audit each result against its original GLB prompt before returning structured JSON.

Singapore interpretation layer

Country context is added dynamically for `SGP`. It recognizes equivalent local structures and terminology, including CPF, statutory versus supplementary benefits, and SGD limits, while explicitly prohibiting inference from general country knowledge when the supplied profiles do not contain evidence.

5. Batching, Parallelism, and Retry



Company and Market are not processed in separate calls. Each model call receives both the complete Company topic section and the complete Market topic section, together with up to 15 datapoint definitions.

For example, 47 Healthcare datapoints are divided into `15 + 15 + 15 + 2`, with the same complete Healthcare evidence supplied to every batch.

Batches execute in parallel under a semaphore to limit concurrency. After each response, requested keys are compared with raw structured output and parsed rows. Missing keys are retried once with the same full topic evidence and only the omitted datapoint definitions. Results are then deduplicated, validated, and restored to authoritative catalog order.

6. Alignment and Validation Improvements

- Deterministic prevalence and percentile rules validate straightforward, like-for-like measures.
- Comparability checks prevent direct numeric comparison across incompatible bases, such as annual versus per-birth limits.
- Compound or qualitative provisions use evidence-grounded holistic alignment rather than the first number found.
- Valid qualitative values such as unlimited or no overall maximum are not treated as numeric extraction failures.
- Currency parsing recognizes SGD/S$ and HKD/HK$ forms.
- Parent-child dependency rules, output formatting, and authoritative source ordering are enforced after aggregation.

Explicit negative normalization

A post-run review found that `Not offered` had been treated as an affirmative substantive value. The normalization was corrected and regression tested. Expected outcomes now include:

Company	Market	Correct alignment
Not offered	0% of peers offer benefit	= Broadly Aligned (0.0%)
Not offered	66.7% of peers offer benefit	-- Significantly Below (66.7%)

7. Model Configuration

Model family	Request configuration	Purpose
GPT-5 / Luna / Terra	Reasoning effort; no temperature or top-p	Avoid unsupported-parameter failures and support reasoning-model evaluation.
GPT-4.1	Temperature and optional top-p retained	Enable controlled tests at temperature 0 and 0.2.

The latest evaluated deployment used GPT-5.6 Luna with reasoning effort `low`. Terra was not available in the Foundry deployment inventory at the time of testing.

8. Latest Singapore Run

Measure	Result
Catalog	GLB V1 Standard, 160 authoritative keys
Model	GPT-5.6 Luna, low reasoning

Measure	Result
Rows returned	160
Passed / corrected / review	124 / 14 / 22
Missing / failed	0 / 0
Customer-safe / review-required	138 / 22
Internal batches	15
Initially omitted / recovered	1 / 1
Parser-dropped keys	0
Missing before placeholder fill	0
Aggregate API-reported tokens	232,367 across initial and retry calls

Token usage is the sum of `response.usage.total_tokens` reported by the model API for all calls. It is not, by itself, an Azure cost figure; billing analysis requires the applicable deployment pricing and token categories.

9. Comparison With the Earlier GPT-4.1 Run

Measure	Earlier GPT-4.1 run	Latest Luna run
Authoritative GLB keys represented	160	160
Missing placeholders	24, all Healthcare	0
Bounded 15-key batching	No	Yes
Omitted-key retry	No	Yes
Model/parser/aggregation diagnostics	Limited	Explicit

Interpretation: Luna shows better completeness than the earlier GPT-4.1 run, but this is not a controlled model-only comparison. The Luna run also benefited from batching, retries, lossless catalog JSON, and stronger extraction instructions. Factual superiority has not yet been established.

10. Known Limitation

Structural completeness does not guarantee factual correctness. In the latest run, the datapoint `risk_benefits_disability_short_term_std_including_sick_leave_prevalence` was returned with Market Practice `Not Provided`, although the Risk Benefits section explicitly states Short-Term Disability prevalence of 0.0%. Because the key was present, omitted-key retry did not activate.

This result is valid as a progress build and as evidence that missing-key handling works. It is not yet a final factual-accuracy benchmark.

11. Verification Completed

- VM validation suite: 12 tests passed.
- Expanded local suite, including explicit negative regression coverage: 13 tests passed.
- Container validation confirmed the GLB V1 catalog, 160 unique keys, and corrected negative normalization.
- Foundry deployment completed successfully with the immutable corrected image.
- End-to-end invocation completed with all 160 rows and no parser or placeholder losses.

12. Recommended Next Steps

1. Add evidence-aware retry for semantically incomplete values such as `Not Provided`, while preventing unsupported replacement.
2. Score Company and Market extraction against the golden dataset by fact, characteristic, statistic, population, and denominator, not wording similarity.
3. Run GPT-4.1 at temperature 0 and 0.2 through the identical updated pipeline.

4. Repeat selected Luna and GPT-4.1 runs to measure consistency.
5. Evaluate Terra on a representative subset when a deployment is available, then expand only if quality justifies cost and latency.
6. Generate and test the Simplified V2 catalog using the same inputs and model after the V1 factual baseline is established.

This report summarizes implementation and validation completed for the Singapore Benefits Review proof of concept. Alignment metrics should be interpreted only after factual Company and Market extraction has been validated.